北京郵電大學 Queen Mary University of London

EBU5408 Digital Audio Fundamentals

Lab 1 – April 2026

This lab is a guided exploration of the characteristics of speech and music sounds. It contains two main exercises.

The *tasks* of Exercise 1 use Audacity; the *tasks* of Exercise 2 use MATLAB.

Learning Objectives
By the end of this lab, you will be able to:

- Load and visualise speech and music signals
- Use spectrograms to observe time–frequency differences
- Understand why speech and music require different processing techniques

All Lab *deliverables* must be submitted to QM+ BEFORE the lab session.

Lab attendance is mandatory. During the lab: interact with your TA and answer their questions about your work (15%).

Lab 1 Submission to QM+ (10%):

- Save in a "results folder" all the lab deliverables.
- Name your folder: 'Lab1_EBU5408_xxxxxxxx' where xxxxxxxx is your QM student number.
- **Upload your folder as a zip archive to QM+ in the EBU5408 course area by Friday April 3rd (firm deadline).**

No submission of the Lab deliverables will be accepted after this deadline.

No submission of the Lab deliverables will result in a mark of 0 for Lab 1.

IMPORTANT: Plagiarism (copying from other students or copying the work of others without proper referencing) is cheating and **will not be tolerated**. **IF TWO "FOLDERS" ARE FOUND TO CONTAIN IDENTICAL MATERIAL, BOTH WILL BE GIVEN A MARK OF ZERO.**

The coursework *report* is to be submitted separately, after Lab 2.

Exercise 1. Comparing music and speech signals using Audacity

The objectives of this exercise are to be able to:

- Visually and perceptually compare **speech and music signals**
- Interpret **waveforms and spectrograms**
- Identify key differences in **time structure, frequency content, and harmonics**

A. <u>Importing Audio Signals and Time-Domain Analysis (Waveforms)</u>

1. Open Audacity.
2. Import the provided audio files: speech.wav, and music.wav.
    - *File → Import → Audio*
3. Listen carefully to each signal and observe the waveform display.

- D1: Screenshot showing the two imported tracks **(1 mark)**
- D2: Saved Audacity project **(1 mark)**

- Answer Lab 1 questions **Q1** and **Q2** in the CW report.

B. <u>Spectrogram Analysis</u>

1. Switch each track to **Spectrogram view**:
    - Right click on track name → *Spectrogram*
2. In the spectrogram settings, try different window sizes.
3. Observe time–frequency structure.

- D3: Spectrogram screenshots for speech and music **(1 mark)**

- Answer Lab 1 questions **Q3** and **Q4** in the CW report.

C. Identifying Harmonics and Pitch

1. Select a **voiced segment** in the speech signal.
2. Select a **steady musical note** in the music signal.
3. Use the spectrogram to:
    - Identify harmonic spacing
    - Estimate fundamental frequency

Hint: In the spectrogram settings, try different Max Frequencies.

- D4: Spectrogram screenshots of the selected segment/note **(1 mark)**

- Answer Lab 1 question **Q5** in the CW report.

Exercise 2 – Creating and Analysing Spectrograms of Audio Signals Using MATLAB

The objective of this exercise is to generate and compare time–frequency representations of different audio signals (noise, music, and speech) using spectrograms created in MATLAB, and to relate visual patterns to signal characteristics. It also explores how speech sounds and speaker characteristics (such as pitch and gender) are represented in spectrograms.

A. Create a MATLAB function named mySpectrogram.m

Create a MATLAB function, which can:

- Read an audio signal
- Ensure the signal is mono
- Generate and display a spectrogram using MATLAB's spectrogram function
- Display time (seconds) on the X-axis and frequency (Hz) on the Y-axis

You should use the following version of the spectrogram command:

```
spectrogram(y, w, noverlap, nfft, fs, 'yaxis');
```

Where:

- y is the mono time-domain signal
- fs is the sampling frequency
- nfft corresponds to a 20 ms analysis window
- noverlap < nfft
- w is a Hamming window defined as: *w = 0.54 - 0.46 * cos(2*pi*(0:nfft-1)/(nfft-1))*;

Hint: For a sampling rate fs, a 20 ms window corresponds to nfft = round (0.02 * fs);

Ensure that:

- The spectrogram axes are correctly labelled
- The X axis shows time and the Y axis shows frequencies.
- The plot is clearly readable

- D5: Your MATLAB **mySpectrogram.m** code file **(1 mark)**

B. Spectrograms of different signal types

Using your mySpectrogram function, generate spectrograms for the following audio files:

- tv-static-noise.mp3
- viola.wav
- sentence.wav

For consistent comparison:

- When necessary, resample signals to 8 kHz using MATLAB's resample function
- Ensure all spectrograms:
  - Have a frequency range from 0 to 4 kHz
  - Are displayed in the same figure size, for example:
    `figure('Position', [200 250 800 250]);`
- Include the filename as the plot title

➢ D6: Each spectrogram as a separate .jpg image **(1 mark)**

- Answer Lab 1 question **Q6** in the CW report.

C. Recording of a speech signal and spectrogram of a single speaker

1. Record a short spoken sentence **in English** using your own voice, stating your name and favourite food. For example: "*My name is ... and my favourite food is ...*"
2. Using the mySpectrogram function you developed, plot the spectrogram of your recorded speech signal.

Hint: To better observe pitch-related features, you may want to increase the figure size or zoom into the frequency axis, for example: `figure('Position', [200 250 800 500]);`

➢ D7: The recording as a mono audio file **(1 mark)**
➢ D8: The spectrogram as a .jpg image **(1 mark)**

- Answer Lab 1 question **Q7** in the CW report.

D. <u>Multiple speakers and comparative spectrogram analysis of voice diversity</u>

1. Create **three recordings of the same short sentence of your choice**, spoken by yourself and two other speakers, including **both male and female voices.** Each recording should be between 10 and 20 seconds. Ensure similar recording conditions (same device, quiet environment).
2. Plot the spectrogram of each of the three speech recordings using mySpectrogram.

- D9: All recordings **(1 mark)**
- D10: Each spectrogram saved as a separate .jpg image **(1 mark)**

- Answer Lab 1 question **Q8** in the CW report.